SPE-229813-MS

Comprehensive Long Term Production Forecast Workflow in Giant Tight Gas Field Development Through Various Prediction and Ensemble Modeling Methodologies

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This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

Abstract

Khazzan Field is a giant tight gas-condensate field located in Oman with a total block area of around 2000 sq.km. Following a gas discovery in 1994, bp entered the block in 2007 and subsequently sanctioned the Khazzan tight gas-condensate development. The longterm full field production forecasting is challenging to appropriately represent the reservoir behavior in relatively wide area coverage. Therefore, a unique approach was taken using various reservoir engineering methodologies combining Rate Transient Analysis (RTA), numerical and analytical models coupled with the bp's propriety Top-Down Reservoir Modelling (TDRM™, Williams et.al, to understand the impact against the uncertainties and to generate representative long term production profiles.

RTA is performed to estimate connected volume for each producing well. Due to the conservative nature of RTA in tight reservoirs in early production history, uncertainty analysis is done for every well to evaluate the well potential using numerical and analytical models coupled with TDRM™. Sector models were also built to model the resources from less mature areas in the field.

Using the above approach, probabilistic resource distributions were established for almost every current or future wellwhich were then combined through a novel methodology to generate full field resource distribution honoring dependency and independency of various blocks. Deterministic scenarios from these distributions were fed into the integrated asset modelling (IAM) toolkit to generate full field production profiles.

Traditional approach for generating simulation models in tight reservoir with wells that are hydraulically fractured might generate very complicated models that would be impractical to use. Using various reservoir engineering methodologies and different toolkits coupled with uncertainty analysis as described above enabled evaluating each part of the field separately and then generate full field resource distribution. This approach provided robust production profiles and a clearer understanding of resources, supporting effective business planning and value optimization under uncertainty.

Introduction

Khazzan Field in located around 350 kilometers southwest of the capital Muscat. It is currently producing 1.5 Bscf/d which is third the production of Oman. Primary reservoir in Khazzan Field is Barik formation which is tight gas condensate reservoir. The secondary reservoir is Miqrat which is even more complicated. There are around 150 producing wells in the field and 10s of wells still to be drilled. All wells are hydraulically fractured to improve the productivity of the well.

The level of heterogeneity is relatively high in Barik and Miqrat formations laterally and vertically. The quality of the reservoir deteriorates from west to east and from south to north. The permeability ranges between 0.1 to 40 mD, gas-condensate ratio (CGR) from 5 to 100 stb/mmscf. Given the complexity of the reservoir and the fact that wells have hydraulic fractures, developing full field model for such highly heterogenies reservoir is challenging. Capturing all physical phenomena in the reservoir would require highly complex models with substantial computational demands, making them impractical. To address this, bp Oman applied various modelling approaches combined with uncertainty analysis to predict the performance of the field and generate full field profiles.

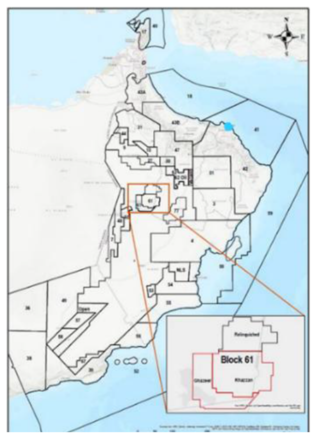

Figure 1—Block61 Location in Oman

Modelling Strategy

Modelling strategy is essential to define the most appropriate modelling methodologies and tools that are able to answer the business questions while addressing the relevant uncertainties. In Khazzan Field, comprehensive modelling strategy was generated aiming to define the most practical modelling methodologies that accounts for subsurface and surface gas molecule interactions, can assist with infill decisions, evaluate condensate banking issues, and predict current and future partial depletion in multi-layer heterogeneous geological settings, and generates probabilistic profiles for the entire field. An evaluation was conducted to define what is ideal and what is practical to address the previously mentioned objectives as following:

What are the expectations?

Ideal:

A comprehensive model that accounts for subsurface and surface gas molecule interactions can assist with infill decisions, evaluate condensate banking issues, and predict current and future partial depletion in multi-layer heterogeneous geological settings, and generates probabilistic profiles for entire fields.

Practical:

Develop models to account for surface well interactions affecting WHP, while using sector models to study subsurface interactions. This includes analyzing infill effects, etc., and scaling insights to the entire field. Segment the field into different sections and strategic planning element (SPE), create probabilistic profiles for each, and integrate these profiles using Monte Carlo methods and surface network models.

What does it take (effort)?

Ideal:

The full-field compositional model spans an area of more than 1000 acres and encompasses 100s of wells. It incorporates fine grid resolution both vertically and laterally and includes local grid refinement near the wellbore to accurately represent fractures. Additionally, it includes a surface network that connects 100s of wells to the gathering system via hundreds of kilometers of pipelines.

*Practical***:**

- Utilization of tools such as RTA, analytical models, and sector models, including single well model (SWM)
- Each well is represented by an SWM (analytical model or simulation) and connected to other wells through the IAM.
- Each well has its own probabilistic profiles. Field probabilistic profiles are created using the Monte Carlo approach (internal code).

The objectives that are outlined above can be addressed by various modelling techniques. The approach conducted is Khazzan Field aimed to generate the most practical approach that is fit for purpose with decent amount of effort with the following steps:

1. Start with business questions: what are the objectives of modelling
2. Mapped out how each tool can address these business questions: most appropriate modelling tool for each objective
3. Make sure all elements of modeling strategy executed: proper plan and timeline for each element in modelling strategy

Modelling Workflow for Full Field Production Forecast

The field is divided into different Strategic Planning Elements (SPEs): Barik Main Development Area (including existing and future wells), Miqrat Main Development, Miqrat Infills, Barik Infills, KZN NE, GZR West, and GZR SW. Probabilistic profiles are created for each SPE using RTA analytical models for existing wells, Nexus sector models for infill wells, and analytical models for future wells. Multiple production scenarios are generated using Monte Carlo methods, which combine resource distributions from SPEs. Deterministic full field profiles are then produced using GAP (from Petex IAM toolkit) to simulate the surface network. The steps of this workflow are described below.

Rate Transient Analysis (RTA) Workflow

RTA is a primary reservoir engineering methodology heavily utilized in block-61 to assess well and reservoir performance and quantify dynamically connected gas in place (Ghanbarzadeh et. al.). RTA is particularly useful in unconventional and tight formations where obtaining connected volumes with other reservoir engineering methods, such as PTA or P/z analysis, is less practical due to extended wells shut-in requirements that can lead to significant production deferrals. RTA utilizes production data and flowing bottom-hole pressure to estimate gas in place without necessitating well shut-ins. While RTA is a valuable tool, it has limitations similar to other reservoir engineering approaches. Specifically, RTA is a single-phase solution and does not consider multi-phase effects that may occur in the reservoir. It also tends to be conservative when wells have a short production history. In tight formations, wells often remain in the transient flow regime for extended periods, and boundary dominated flow (BDF) typically develops only

after several years of production. Additionally, RTA relies on graphical solutions, which may introduce subjectivity into interpretation, especially since real data can be noisy.
bp developed sophisticated workflow to generate robust RTA interpretation while accounting for limitations and uncertainty around the interpretation. In this workflow, Flowing material balance (FMB) is used to estimate connected gas in place. Then various type curves are utilized to mainly inform whether the wells have seen BDF or still in transient flow in addition to providing estimation of connected gas in place along with formation permeability, fracture half length (xf) and fracture conductivity (F_CD). Type curves that are used are Blasingame with finite conductivity frac, Blasingame with infinite conductive fracture and NPI. Estimation of connected gas in place from type curves is crossed checked with FMB in order to make sure common interpretation is established. Then interpretation from FMB and type curves are fed into analytical model to check the quality of match to actual historical data. Feedback loop in between FMB, type curves and analytical model is performed until common interpretation is achieved that provide accepted match on all methodologies and to actual historical production data as in Figure 2. RTA workflow is discussed in much greater detail by Al Hinai and Motealleh

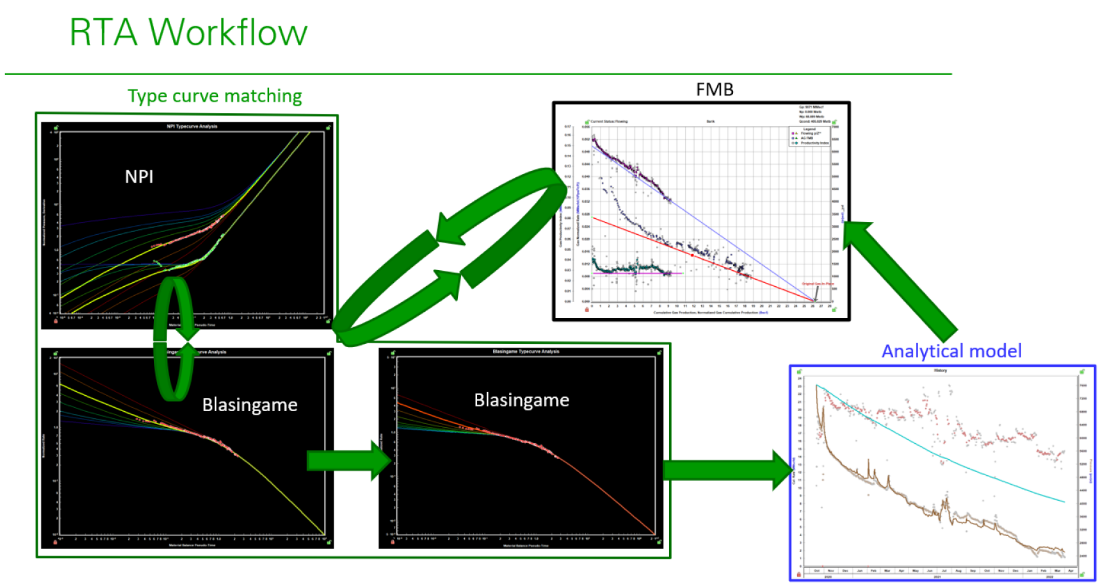

Figure 2—RTA workflow impeded in Block-61 to generate robust interpretation. Iterations between FMB, various type curves and analytical models to generate interpretations that achieve accepted match in all methodologies.

Every producing wells in the block has RTA models which then will be utilized to generate life of field profile for every well.

Finite Difference Multi-phase Reservoir Simulation: Sector Models

While RTA is used to mainly understand well by well dynamic connected GIP and predict future performance based on that, sophisticated three numerical models were built mainly for the following objectives:

- Value of infill wells
- Development phase sequence optimization: Compression vs velocity string

Key uncertainties are considered to be factors influencing reservoir heterogeneity; therefore, the emphasis is on representing a range of geological scenarios away from well control rather than focusing on near-wellbore phenomena such as condensate banking or detailed fracture modelling. Calibration primarily involves matching each well's connected volume and RTA results.

The models use grid dimensions of 200m × 200m × 0.3m with high vertical resolution. Fluids are characterized in 18 black oil PVT regions. An Equation of State (EoS) was developed to replicate fluid behavior in the Barik formation. Eighteen PVT regions were determined to be the minimum required to simulate fluid behaviors comparable to those in a compositional model. Representation of hydraulic fractures in these models is simplified. Local grid refinement is not applied; instead, permeability modifiers are introduced in cells identified as likely fracture propagation paths. This approach was validated by comparing results to those obtained with local grid refinement, showing consistent matches in gas and condensate rates and FBHP, which was deemed sufficient for the intended purposes. The replacement of the compositional model with a black oil model, along with the removal of LGR, significantly reduces CPU run time, facilitating the assessment of numerous uncertainties and adding analytical value.

A distinct methodology was used to calibrate the models to historical data, with a primary goal of matching connected gas in place as indicated by RTA. Dedicated tools were developed to support this objective. Further details on this approach are beyond the scope of this paper.

Following are the key outcome of these models:

Key learnings:

- Understanding of reservoir continuity.
  - Testing different geological scenarios and dynamic data calibration drive towards higher reservoir connectivity cases in the ensemble.
- Model results aligned with expected rock type contribution.
  - No contribution from poro<4% rock. 4%<Poro<8% contribution is uncertain and both scenarios included in the ensemble
  - Future surveillance needed to understand flow contribution from poorer rock quality.
- Impact of uncertainties on value of infill range
  - Higher the connectivity, lower infill incremental (assuming infills targeting Middle Barik)
- Model EUR/well vs RTA comparison
  - RTA EUR on the conservative side.
  - Well by well deep dive conducted to explain and resolve the differences.
  - RTA interpretation has limitations. Underlying drainage area shape assumptions might cause RTA results to show boundary dominated flow too early.
  - Model calibration method is a simplified version of RTA connected volume formulation, rather than fully replicating it. Therefore, model results represent an approximate connected volume estimate.

Existing wells that are within the area covered by sector models now have performance forecast based on RTA and another one based on sector models. In general, RTA is conservative when compared to sector models which is expected as RTA is only able to detect the volume from rocks that contribute at the time of analysis. Where numerical models predict that poorer rocks do contribute when they are exposed to higher depletions. Utilizing results of RTA and learnings from sector models which then upscaled to the entire field, Monte Carlo probabilistic run is performed for every well to generate distribution of estimated ultimate recovery (EUR) as in Figure 3 . Black tank un the figure is defined by RTA, high case is defined from numerical models and low case is defined based on possible error in RTA. As illustrated in the plot, the range is not very wide as all distributions are anchored to RTA

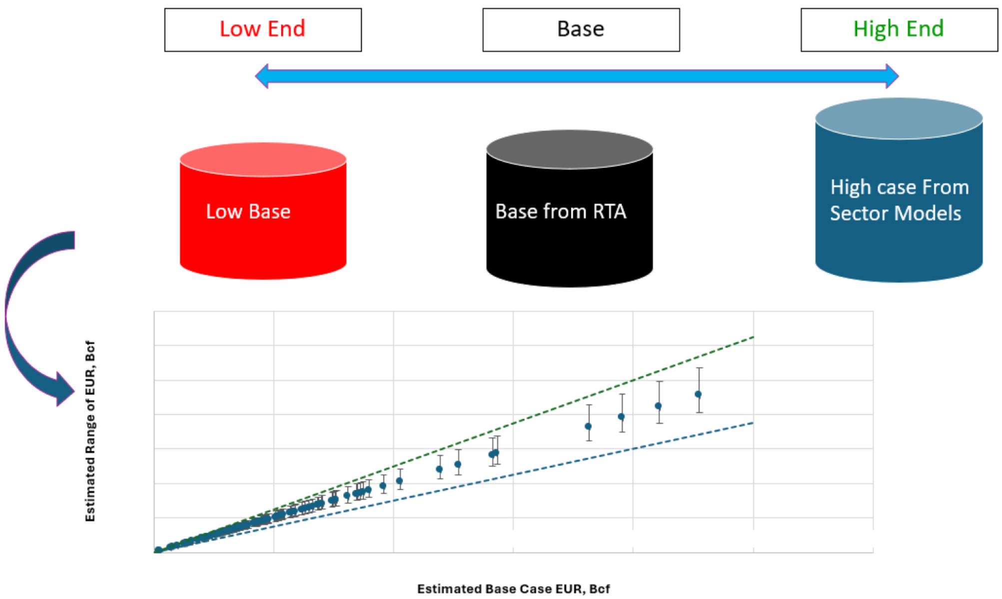

Figure 3—Probabilistic profiles utilizing outcome from RTA and learnings from sector model

One main objective of the sector models is assessing the value of infills. Key learning from sector models is that value of infills in inversely correlated with improvement in reservoir connectivity. If the reservoir is well connected in all layers then value of infills will be less where volumes will be drained by main development wells. Distribution of recovery factor (RF) uplift by infills is generated from sector models. This learning is then taken and upscaled to entire field. RTA work coupled with sector models will generate resource distribution for existing wells and potential infilling wells.

Analytical Models

Conducting TDRM™ analyses for hundreds of wells in a complex reservoir and large field such as Block-61 presents significant challenges and can require substantial time and effort using conventional methods. To address this, bp Oman has developed an in-house analytical model tailored to type wells in the Khazzan field. The model is grounded in established analytical solutions from literature for linear, radial, and boundary-dominated flow regimes, along with expressions from numerical models to transition between flows. While details of the model fall outside the scope of this paper, they will be discussed in a future publication. Uncertainty analysis is subsequently performed by varying several uncertainty variables, yielding resource distributions for each potential well in the field. By integrating the in-house analytical model with TDRM™, it is now feasible to conduct uncertainty analyses for hundreds of scenarios within minutes. As presented in Figure 4, all wells across the investigated areas display their respective resource distributions and profile clouds.

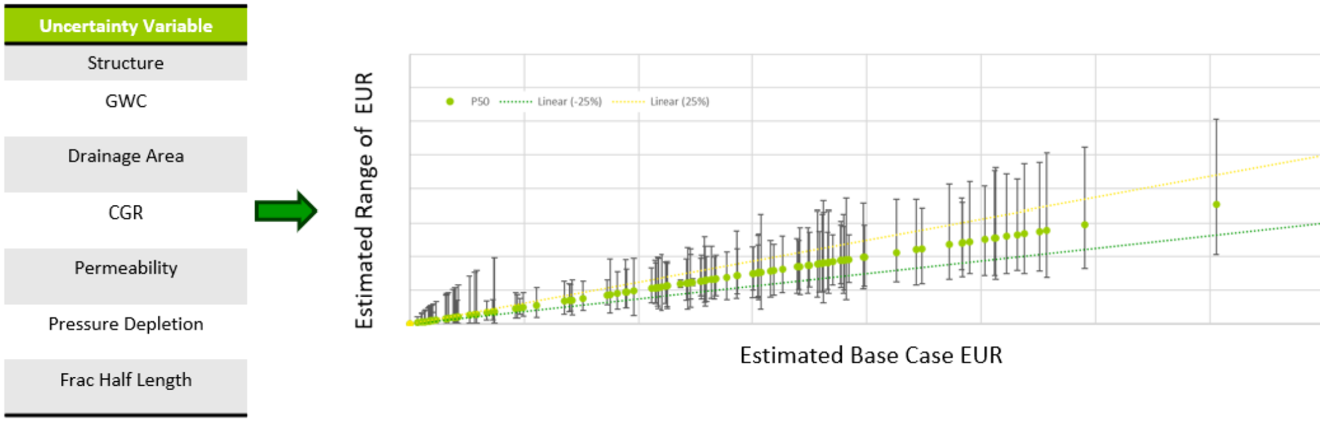

Figure 4—Every possible well in the field has distribution of resources. The uncertainty in unmatured areas is higher which is represented by wide range of resource

Probabilistic Combination: Monte Carlo Approach for Full Field Potential

Each strategic planning element (SPE) is modelled independently, resulting in separate resource distributions for each SPE within the field. The EUR distribution for each SPE is described by three values: P90 (pessimistic case), mean (expected case), and P10 (optimistic case). These values represent the range of possible outcomes for each SPE and offer a broad perspective on potential production scenarios. To construct a full-field distribution profile, these three points are used for each SPE. The distribution profiles from all SPEs are combined using Monte Carlo simulation, a statistical method that generates numerous random samples from the distribution profiles to form a complete field distribution. By sampling randomly from the P90, mean, and P10 values of each SPE, the Monte Carlo simulation produces a variety of probable outcomes for the entire field. This process provides insights into overall production potential and associated risks. The result is a distribution that reflects the total production potential for the field. Figure 5 illustrates this methodology.

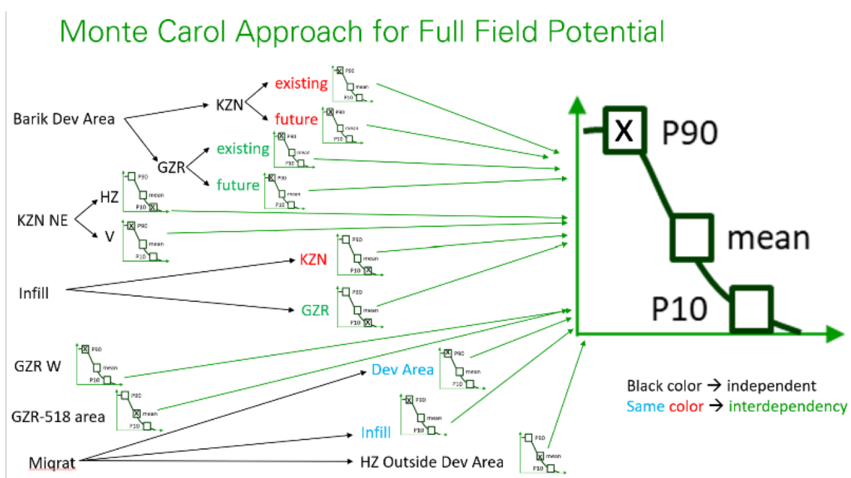

Figure 5—Methodology of combining EUR distribution from every SPE to generate full field potential resource distribution

This distribution can be used to assess various development scenarios and support decision-making processes. The full field distribution enables comprehensive scenario analysis, such as identifying combinations of SPEs and activities that meet production targets under differing conditions (for example, with or without compression). By considering the full range of possible outcomes, decision-makers can prepare for different contingencies, including both geological risks and opportunities. This approach helps ensure that the development plan remains flexible and responsive to changing circumstances. Information derived from the full field distribution informs strategic business decisions, including considerations related to investment in compression or the selection of particular SPEs, supporting both technical and economic viability in the development plan.

Integrated Asset Modelling

An integrated Asset Modelling tool was developed for full field model incorporating all reservoirs, wells, and the pipeline network across the entire field. The model includes all potential wells in the area. This represents the first comprehensive field model constructed for Khazzan Field following the commencement of production, with the aim of generating long-term field profiles and supporting optimization of the business plan, such as well and rig count.

Each well is represented by a single tank utilizing the Tight GAS (TG) MBal model. There is no reservoir interaction between wells; interactions occur only through the surface network. The selection of the tight gas MBal model is based on the following considerations:

- The 650+ km of flow lines necessitates accurate modeling of pressure drops.
- The large number of wells at varying stages of maturity requires modelling potential interactions via the network (back-out).
- The model allows for reasonable run-times when assessing different compression concepts, since coupling full field geocellular models with a surface network would increase run times significantly unless up-scaling is used.
- The southern part of the field has little or no aquifer presence, reducing likelihood of early water breakthrough or substantial pressure support.
- Hydraulic fracturing of wells must be accounted for in the model.

Certain assumptions underlying this approach include:

- Condensate banking effects are not considered.
- The shift of the no flow boundary between wells due to differential depletion is deemed negligible, as the total connected GIIP is unchanged and distant gas has limited impact.

It is assumed that all GIIP is equally accessible to each well, contributing to flow from the outset. While this assumption may not always hold, uncertainty in connected volume is addressed by employing models with variable GIIP.

Deterministic scenarios are selected from the full-field distribution (Figure 5) and input into GAP to create complete field production profiles. This workflow produces low, reference, and high case profiles for the field, which inform the optimization of business planning.

Conclusions

A comprehensive long-term production forecasting workflow was successfully implemented for the development of a major tight gas field, utilizing a range of predictive methodologies including Rate Transient Analysis (RTA), sector reservoir simulation models, probabilistic approaches, and integrated production modeling. The key conclusions from this study are as follows:

- The selection of an appropriate modeling strategy is critical to determine suitable methodologies and tools for addressing business objectives and accommodating relevant uncertainties.
- A flowing material balance via RTA was effectively developed to estimate connected gas in place from current producers and to determine important reservoir properties such as formation permeability, fracture half-length (xf), fracture conductivity (F_CD), and reservoir boundaries.

This analysis incorporates a robust range and assessment of uncertainty based on historical production data.

- Sector reservoir simulation models were employed to calibrate RTA results and to provide insights beyond existing production, including infill well planning and development phase sequencing. Additionally, these models facilitated the evaluation of key uncertainties related to reservoir heterogeneity, enhancing confidence in the representation of geological scenarios beyond areas of established well control.
- The probabilistic combination approach enabled consideration of the full spectrum of potential outcomes. Three representative subsurface scenarios were selected from the ensemble to encompass a broad array of subsurface uncertainties for a detailed assessment of the field development plan.
- Integrated production modeling was applied, incorporating all reservoirs, wells, and the pipeline network across the field, thus increasing confidence in the predictability of system performance given the network configuration.

Acknowledgements

The authors would like to thank bp, OQ, PTTEP, Petronas and the Ministry of Energy and Mineral of Oman for allowing the
publication of this work. Authors also thank the entire Oman subsurface team for their outstanding efforts in generating the workflow detailed in this paper.

Nomenclature

TDRM™ = Top-Down Reservoir Modelling
RTA = Rate transient analysis
Xf = fracture half length
F_CD = Fracture dimensionless conductivity
FMB = Flowing material balance
GIIP = Gas initially in place
PTA = Pressure transient analysis
EoS = Equation of state
PVT = Pressure volume temperature
BHP = Bottom hole pressure
BDF = Boundary dominated flow
IAM = Integrated asset modeling

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